

Jaehwan Jeong

✉ Email | 🎓 Google Scholar | 🔗 LinkedIn | 🏠 Website

RESEARCH INTERESTS

Embodied AI, Robotic Learning, Autonomous Systems, Sim-to-Real
Vision-Language Model, Multi-modal learning, Generative model, AI safety

EDUCATION

- **Korea University** Mar. 2024 - Fe. 2029 (Expected)
Ph.D. - Artificial Intelligence Seoul, South Korea
- **Chung-Ang University** Mar. 2017 - Fe. 2021
B.E. - Electrical & Electronic Engineering Seoul, South Korea

EXPERIENCE

- **Computer Vision Lab @ Korea University** Mar. 2024 - Present
Ph.D. Student Researcher (Advisor: Prof. Sangpil Kim)
 - Optimized 3D occupancy forecasting architectures (collaborated with Dr. Jonghyun Choi @ **HYUNDAI Motors**)
 - Adversarial noise for Deepfake AI safety (collaborated with Dr. Jaewook Chung @ **SAMSUNG Research**) - [C2]
 - Multi-modal audio-to-video generation (collaborated with Dr. Eugenio Culurciello @ **PURDUE Univ.**) - [J1]
- **Structures-Computer Interaction Lab @ University of California, Los Angeles (UCLA)** Mar. 2025 - Mar. 2026
Visiting Graduate Researcher; **3D Farm Project Lead** (Advisor: Prof. M. Khalid Jawed)
 - VLA & RL-driven Sim2Real deployment for mobile robots (collaborated with Dr. Joo @ **NVIDIA Research**) - [C4]
 - 3D Plant skeletonization & Physics-based simulation for robotic pollination - [J2]
 - Robotic agricultural 3D reconstruction benchmark and dataset (collaborated with **NDSU**) - [C3]
- **Computer Vision Lab @ Korea University** Jul. 2023 - Fe. 2024
Undergraduate Research Intern (Advisor: Prof. Sangpil Kim)
 - Diffusion-based long video generation (collaborated with Dr. Byeon @ **NVIDIA Research**) - [C1]
- **Military Officer** Mar. 2021 - Jun. 2023
Signal Company, 5th Armored Brigade, Republic of Korea Army
 - Operations and Tactical Planner - 1st Lieutenant
 - Wired communication network management (UTP, Optical cables) - 2nd Lieutenant

PUBLICATIONS

C=CONFERENCE, J=JOURNAL

- [C4] **Jaehwan Jeong**, Evelyn Zhu, Jinying Lin, Emmanuel Jaimes, Tuan-Anh Vu, Jungseock Joo, Sangpil Kim, M. Khalid Jawed, *Your Vision-Language-Action Model Already Has Attention Heads For Path Deviation Detection*, under review, 2026 [PDF]
- [J2] **Jaehwan Jeong***, Tuan-Anh Vu*, Radha Lahoti, Jiawen Wang, Vivek Alumootil, Sangpil Kim, M. Khalid Jawed, *Vision-Guided Targeted Grasping and Vibration for Robotic Pollination in Controlled Environments*, under review, 2026 [PDF]
- [C3] **Jaehwan Jeong**, Tuan-Anh Vu, Mohammad Jony, Shahab Ahmad, Md. Mukhlesur Rahman, Sangpil Kim, M. Khalid Jawed, *AgriChrono: A Multi-modal Dataset Capturing Crop Growth and Lighting Variability with a Field Robot*, under review, 2026 [PDF] [Project] [Code]
- [C2] **Jaehwan Jeong**, Sumin In, Sieun Kim, Shin han yi, Jonghen Jeong, Sang Ho Yoon, Jaewook Chung, Sangpil Kim, *FaceShield: Defending Facial Image against Deepfake Threats*, International Conference on Computer Vision (ICCV) 2025 [PDF] [Code]
- [J1] Sieun Kim, **Jaehwan Jeong**, Sumin In, Seung Hyun Lee, Seungryong Kim, Saerom Kim, Wooyeol Baek, Sang Ho Yoon, Eugenio Culurciello, Sangpil Kim, *Semantically Complex Audio to Video Generation with Audio Source Separation*, Engineering Applications of Artificial Intelligence (JCR IF Top 10%) 2025 [PDF] [Code]
- [C1] Gyeongrok Oh, **Jaehwan Jeong**, Sieun Kim, Wonmin Byeon, Jinkyu Kim, Sungwoong Kim, Sangpil Kim, *MEVG: Multi-event Video Generation with Text-to-Video Models*, European Conference on Computer Vision (ECCV) 2024 [PDF] [Code]

ACADEMIC SERVICE

- **Conference Reviewer**
CVPR'25-26, ICCV'25, ECCV'26, NeurIPS'26, BMVC'26, DLT'25
- **Journal Reviewer**
EAAI'25-26, CVIU'25

PATENTS

- [P1] Jaehwan Jeong, Sangpil Kim, *Method and Apparatus for Protecting Facial Image based on Disturbance Signal to Counter Deepfake Attack*, Korean Patent, No. 10-2025-0118588 (2025.08.25)

SKILLS

- | | |
|---|---|
| <ul style="list-style-type: none">• AI & Perception
Vision-Language Models, Reinforcement Learning, Adversarial Attack, Diffusion Models, GANs | <ul style="list-style-type: none">• Robotics & Sensor Fusion
Unmanned Ground Vehicle (UGV), Robotic Arms, Simulation, ROS 2, RGB-D Cameras, LiDAR, IMU |
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REFERENCES

- **Assoc. Prof. Sangpil Kim**
Department of Artificial Intelligence, Korea University, South Korea
- **Assoc. Prof. M. Khalid Jawed**
Department of Mechanical and Aerospace Engineering, University of California, Los Angeles
- **Dr. Jungseock Joo**
Senior Engineering Manager @ NVIDIA | ex-Prof @ UCLA
- **Postdoc. Tuan-Anh Vu**
Department of Mechanical and Aerospace Engineering, University of California, Los Angeles